

STOCHASTIC METHODS IN WATER RESOURCES

Unit 4: Time series analysis

Lecture 11: Univariate, linear stochastic models

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Application to non-normal series

Generalities

- ▶ The models discussed previously are assumed to be applied to temporal discrete series, that are transformed to **second-order stationarity**.
- ▶ This implies that the distribution of the **random component** is **normal**.
- ▶ In the case of **annual series**, this approximation would generally suffice.
- ▶ However, modifications may be necessary for coping with **monthly** and/or **daily** series.
- ▶ If the ξ_t is **non-normally distributed** with **skewness** γ_ξ , then the skewness γ_η to be incorporated in the random component η_t depends on the model and its parameters.
- ▶ An alternative procedure is to **transform** observed values so that the distributions are **normal**.

Application to non-normal series

Preservation of skewness in Markov model

The model is of the form

$$\xi_t = \phi_{1,1}\xi_{t-1} + \eta_t \quad (1)$$

where $\phi_{1,1} = \rho_1$ is the **lag-one autocorrelation coefficient**. By cubing both sides and taking expectations

$$E[\xi_t^3] = E[\rho_1^3 \xi_{t-1}^3 + 3\rho_1^2 \xi_{t-1}^2 \eta_t + 3\rho_1 \xi_{t-1} \eta_t^2 + \eta_t^3] \quad (2)$$

Because ξ_{t-1} and η_t are independent, the expectations of the second and third terms on the right are zero. Also, it is known that for an **AR(1)**, $\sigma_\eta^2 = 1 - \rho_1^2$ and $E[\xi_t] = E[\eta_t] = 0$. Therefore

$$\gamma_\eta = \frac{\gamma_\xi(1 - \rho_1^3)}{(1 - \rho_1^2)^{3/2}} \quad (3)$$

in which, for estimation purposes, ρ_1 is replaced by r_1 , its **sample estimate**.

Application to non-normal series

Box-Cox transformations

Box and Cox in 1964 suggest the following method of transforming and observed value x

$$Y = \begin{cases} \frac{x^\lambda - 1}{\lambda} & \lambda \neq 0 \\ \ln(x) & \lambda = 0 \end{cases} \quad (4)$$

The parameter λ could be chosen so that the skewness of the transformed series is a minimum. However, the efficiency of this approach is very low compared with more sophisticated **maximum likelihood** method. A simpler procedure could be for different values of $\lambda = 0, \pm 0.5, \pm 1.0, \dots$ the **sample mean** $\bar{y}$, the **standard deviation** $\hat{\sigma}_y$ (or the more robust interquartile range) and the **median** $\hat{\zeta}_t$ are calculated from the transformed data. Then the statistics $d_\lambda = \frac{\bar{y} - \hat{\zeta}_y}{\hat{\sigma}_y}$ are compared and interpolated if necessary, to find the value of λ for which d_λ is a minimum. A further refinement is to introduce a second parameter by replacing x with $x - \zeta$ say. Of special interest is the situation when $\lambda = 0$ in equation 4. This is the case in which the variable has a **lognormal distribution**.

Application to non-normal series

Data generation using Markov model and lognormal distribution

Consider the transformation $Y = \ln(X - \xi)$ where X has mean μ and variance σ_X^2 ; Y is normally distributed with mean μ_Y and variance σ_Y^2 and ξ is a location parameter, which could be negative; also, the lag-1 autocorrelation coefficients are ρ_I and $\rho_{I,Y}$, respectively.

It can be shown that

$$\mu = e^{\mu_Y + \sigma_Y^2/2} + \xi \quad (5)$$

and

$$\sigma^2 = e^{2\mu_Y + \sigma_Y^2} (e^{\sigma_Y^2} - 1) \quad (6)$$

Also

$$\gamma_1 = (e^{3\sigma_Y^2} - 3e^{\sigma_Y^2} + 2) (e^{\sigma_Y^2} - 1)^{-3/2} \quad (7)$$

where γ_1 is the coefficient of skewness of the population. By substituting sample estimators g_1 , s^2 and $\bar{x}$ in the left-hand side of equations 7, 6 and 5 respectively, estimators s_Y^2 , $\bar{y}$ and $\bar{\xi}$ of the parameters σ_Y^2 , μ_Y and ξ are obtained, in that order. The relationship between ρ_I and $\rho_{I,Y}$ is determined by using equations 5 and 6 as follows

$$E[X_t X_{t+I}] = E[(e^{Y_t} + \xi)(e^{Y_{t+I}} + \xi)] = E[e^{Y_t + Y_{t+I}} + \xi e^{Y_t} + \xi e^{Y_{t+I}} + \xi^2] \quad (8)$$

From equation 5, $E[e^{Y_t}] = e^{\mu_Y + \sigma_Y^2/2}$. It follows that

$$E[e^{Y_t + Y_{t+I}}] = e^{2\mu_Y + \frac{1}{2}\text{var}(Y_t + Y_{t+I})} = e^{2\mu_Y + \sigma_Y^2(1 + \rho_{I,Y})}$$

Application to non-normal series

Data generation using Markov model and lognormal distribution

Therefore, again using equation 5

$$\begin{aligned}\text{cov}(X_t, X_{t+l}) &= E[(X_t - \mu)(X_{t+l} - \mu)] \\ &= E(X_t X_{t+l}) - \mu^2 \\ &= \left[e^{2\mu_Y + \sigma_Y^2(1-\rho_{l,Y})} + 2\xi e^{\mu_Y + \sigma_Y^2/2} + \xi^2 \right] - \left(e^{\mu_Y + \sigma_Y^2/2} + \xi \right)^2 \\ &= e^{2\mu_Y + \sigma_Y^2(1+\rho_{l,Y})} - e^{2\mu_Y + \sigma_Y^2}\end{aligned}$$

Hence, by using equation 6

$$\rho_l = \frac{\text{cov}(X_t, X_{t+l})}{\sigma^2} = \left(e^{\sigma_Y^2 \rho_{l,Y}} - 1 \right) \left(e^{\sigma_Y^2} - 1 \right)^{-1} \quad (9)$$

By substituting the (estimated) lag-one serial correlation coefficient r_1 for ρ_1 in equation 9, an estimator $r_{1,Y}$ of $\rho_{1,Y}$ is obtained. The generating equation for the **Markov lognormal model** is

$$X_t = \xi + \left[e^{\mu_Y(1-\rho_{1,Y})} \right] (X_{t-1} - \xi)^{\rho_{1,Y}} e^{\sigma_Y \eta_t} \quad (10)$$

in which η_t is an independent standard normal variate with a zero **mean** and a **variance** $1 - \rho_{1,Y}^2$. An alternative and more direct method is to use the second of the **Box and Cox transformations** $y = \log(x)$. Then the parameter of, say, the **Markov model** are estimated from the sequence y , and, finally, antilogarithms are taken on the generated data. However, this method has two disadvantages: firstly, **zero values** cannot be transformed, and for this reason an arbitrary small value is sometimes added to all observed values; secondly, statistical and other properties of generated data may differ significantly from the properties of historical data. On the other hand, all values produced will be positive, unlike in the application of equation 10.

Application to non-normal series

Dealing with non-normal distributions in general

- ▶ For application purposes, the differences in **skewness** of the **stochastic component** ξ_t and the **random component** η_t , as given by equation 3 for the **Markov model**, should be noted.
- ▶ A practical method is to calculate the **skewness** and other distribution properties from the estimated independent **residuals** which for a **Markov model** is given by $\tilde{\eta}_t = \tilde{\xi}_t - r_1 \tilde{\xi}_{t-1}$, where $\tilde{\xi}_t$ is an observed standardized variable.
- ▶ This can be extended to the more complicated models by using appropriate forms of $\tilde{\eta}_t$, as given by equations 28, 29 and 30 in the lecture 12.
- ▶ However, it should be borne in mind that that all these procedures are subject to **sampling errors**. Here, **distributions** are deduced from estimated independent **residuals** which may be distorted from model **errors** and **bias** in estimated parameters.
- ▶ The **uncertainties** may be resolved to some extend by tests on generated data.

Generalities

In hydrology, seasonal models applicable to monthly series were originally used by Thomas and Fiering in 1962. Let $x_{t,\tau}$ represent the flow in month t , where $t, t = 1, 2, \dots, 12N$ (and N is the number of years), corresponds to the calendar month $\tau, \tau = 1, 2, \dots, 12$. Then, seasonality is preserved by equating the standardised variable ξ_t used, for example, in $\xi_t = \sum_{i=1}^p \phi_{p,i} \xi_{t-i} + \eta_t$ to the residuals $(x_{t,\tau} - \mu_\tau)/\sigma_\tau$, in which μ_τ and σ_τ represent the **mean** and the **standard deviation** respectively for the calendar month τ . This method is known as **prewhitening**.

Autoregressive seasonal models

The p th-order autoregressive seasonal model takes the form

$$X_{t,\tau} = \mu_\tau + \sigma_\tau \sum_{i=1}^p \frac{\phi_{p,i}(X_{t-i,\tau-i} - \mu_{\tau-i})}{\sigma_{\tau-i}} + \sigma_\tau \eta_t \quad (11)$$

in which $E[\eta_t] = E[\eta_t \eta_{t-k}] = 0$ for $k \neq 0$ and $\text{var}[\eta_t] = E[\eta_t^2] = 1 - R^2$; note that, if $\tau - i \leq 0$, $\tau - i$ should be replaced by $\tau - i + 12$. The order p and the autoregressive parameter $\phi_{p,i}$ are estimated from the standardised ξ_t series.

Because μ_τ and σ_τ have to be estimated from short historical samples, harmonic fitted statistics $\hat{\mu}_\tau$ and $\hat{\sigma}_\tau$ have been substituted in equation 11 for estimating the autoregressive parameters. However, the resulting $\tilde{\xi}_t$ series and hence the autoregressive parameters may be significantly affected by sampling errors. This could happen when the **sample statistics** m_τ and s_τ are used or when their harmonic fitted estimates $\hat{\mu}_\tau$ and $\hat{\sigma}_\tau$ are substituted for the unknown parameters μ_τ and σ_τ . The advantage in the harmonic model is that it needs fewer parameters; also these are less affected by sample size. A further generalization of this model is possible by making $\phi_{p,i}$ seasonal but usually the differences between the seasonal parameters are not significant.

Thomas-Fiering seasonal model

This takes the form

$$X_{t,\tau} = \frac{\mu_\tau + \sigma_\tau \rho_\tau (X_{t-1,\tau-1} - \mu_{\tau-1})}{\sigma_{\tau-1}} + \sigma_\tau \eta_t \quad (12)$$

in which

$$\rho_\tau = \frac{E[(X_{t,\tau} - \mu_\tau)(X_{t-1,\tau-1} - \mu_{\tau-1})]}{\sigma_\tau \sigma_{\tau-1}}$$

is the coefficient of correlation between the flows in months τ and $\tau - 1$. For the random component $E[\eta_t] = E[\eta_t \eta_{t-k}] = 0$ for $k \neq 0$. Also, recall that $\text{var}[\eta_t] = 1 - \rho_\tau^2$.

If **skewness** is to be maintained, then by using a similar derivation as for the equation 3, the **coefficient of skewness** to be applied to the random component in month τ is given by

$$\gamma_{\eta,\tau} = \frac{\gamma_\tau - \rho_\tau^3 \gamma_{\tau-1}}{(1 - \rho_\tau^2)^{3/2}} \quad (13)$$

where γ_τ is the **coefficient of skewness** in $X_{t,\tau}$, $t = 1, 2, 3, \dots$.

The drawback here is that with short records significant differences in the ρ_τ values may not be found. Besides, ρ_τ could be zero for particular months of the year, partly on account of **random climatic movements** within the year.

Seasonal models

Box-Jenkins seasonal model

An **ARIMA** type model can also be used for generating data in which seasonal effects are accounted for. **Backshift** operators such as $B^s \xi_t = \xi_{t-s}$ are used for this purpose when events s time units apart are indetically distributed; for example, if the annual cycle is to be represented through a monthly series, $s = 12$. Consider initially a model of the type **ARIMA(1,0,1)** which is, of course, equivalent to ot **ARMA(1,1)**. It is given by $(1 - \Phi B^{12})\xi_t = (1 - \Theta B^{12})\psi_t$, that is, $\xi_t - \Phi \xi_{t-12} = \psi_t - \Theta \psi_{t-12}$ where Φ and Θ are parameters and ψ_t is a standardised residual. The next step is to apply a difference operator which is given by

$$\begin{aligned}\nabla_s \xi_t &= (1 - B^s)\xi_t \\ &= \xi_t - \xi_{t-s} \\ \nabla_s^2 \xi_t &= (1 - B^s)^2 \xi_t \\ &= \xi_t - 2\xi_{t-s} + \xi_{t-2s}\end{aligned}$$

and in general, $\nabla_s^D \xi_t = (1 - B^s)^D \xi_t$. Accordingly, the seasonal **ARIMA(P,D,Q)_s** model is represented by

$$\Phi_P(B^s)\nabla_s^D \xi_t = \Theta_Q(B^s)\psi_t \quad (14)$$

However, the **residual** $\psi(t) = \Theta_Q^{-1}(B^s)\Phi_P(B^s)\nabla_s^D \xi_t$ is usually not independent, mainly because the **short-lag autocorrelation structure** is not incorporated in the model. For this reason, it is necessary to formulate a multiplicative seasonal model which, in the general case, is the combination of two **ARIMA** types and is denoted by $(p, d, q) \times (P, D, Q)_s$.

Seasonal models

Box-Jenkins seasonal model

The complete **seasonal model** then takes the form

$$\Phi_P(B^s)\phi_P(B)\nabla_s^D\nabla^d\xi_t = \Theta_Q(B^s)\theta_q(B)\eta_t \quad (15)$$

where η_t is a standardised independent variable.

For example, a multiplicative **ARIMA(2,0,0)×(1,1,1)₁₂** model (applied to monthly data) is given by $(1 - \phi_1 B - \phi_2 B^2)(1 - \Phi_1 B^{12})(1 - B^{12})\xi_t = (1 - \Theta_1 B^{12})\eta_t$. In application, the procedure is

1. Obtain a differenced sequence $\nabla_{12}\tilde{\xi}_t = \tilde{\xi}_t - \tilde{\xi}_{t-12}$ from a sample of observed data $\tilde{\xi}_t$, $t = 1, 2, 3, \dots, N$
2. The estimates $\hat{\Phi}_1$ and $\hat{\Theta}_1$ are found by applying a seasonal **ARMA(1,1)₁₂** model to the differenced sequence.
3. An **AR(2)** model is applied to the residuals from the seasonal model. The generating equation, found by expanding the product $(1 - \phi_1 B - \phi_2 B^2)(1 - \Phi_1 B^{12})(1 - B^{12})$ and substituting for the B terms, is as follows

$$\begin{aligned}\xi_t = & \phi_1 \xi_{t-1} + \phi_2 \xi_{t-2} + (1 + \Phi_1) \xi_{t-12} - (\phi_1 + \phi_1 \Phi_1) \xi_{t-13} \\ & - (\phi_2 + \phi_2 \Phi_1) \xi_{t-14} - \Phi_1 \xi_{t-24} + \phi_1 \Phi_1 \xi_{t-25} + \phi_2 \Phi_1 \xi_{t-26} \\ & + \eta_t - \Theta_1 \eta_{t-12}\end{aligned}$$

This type of model is applicable, strictly speaking, only when the **monthly standard derivation** are **not seasonal**.

Seasonal models

Models for daily flows: problems and references

- ▶ When applied to **daily flows** the **autoregressive** and general **Box-Jenkins** models described so far have limitations, which could be serious.
- ▶ One reason is that, because **daily data** have highly **non-normal distributions**, extreme values and properties such as **runs** (a sequence of events with reference to a given magnitude or level which are above the level or alternatively all are below the level) and **crossings** (of the level occurs at the commencement as well as the end of the run) are not maintained through models that were devised originally for **normal processes**.
- ▶ Of equal significance is the fact that the **rapid ascensions** and **slow recessions** that are associated with individual rainfall events are not specifically modelled.
- ▶ A particular method of preserving such time-irreversible characteristics is through **filtered Poisson processes**.
- ▶ In application, a continuous series $X(t)$ of **river flows** is considered to be generated by a number $N(t)$ of "**rainfall shots**" Y_m in the interval $(-\infty, t)$; $N(t)$ is a **Poisson process** with mean ν .
- ▶ In effect, the flow at time t is contributed by pulses $Y_m e^{-b(t-\tau_m)}$, $m = 0 \pm 1 \pm, \dots$, which are zero for $t < \tau_m$.
- ▶ Here, the magnitude of a **shot** Y_m is an **independent exponentially distributed** random variable with **mean** θ , b is the **decay (or hydrograph recession) rate** and τ_m denotes **random arrivals times** of shots which are analogous to the times that **rain storm start**.

Models for daily flows: problems and references

- ▶ The variable m is considered to extend backwards to $-\infty$, so that

$$X(t) = \sum_{m=-\infty}^{N(t)} Y_m e^{-b(t-\tau_m)} \quad (16)$$

- ▶ The model parameters ν , θ , and b in the model are estimated from the **means**, the **standard deviations** and the **lag-one serial correlation** in the data for each month.
- ▶ The integration of equation 17 between limits t and $t - 1$ gives the following discrete process

$$X_t = e^{-bs} X_{t-s} + \eta_t \quad (17)$$

which has a **Gamma density function**.

- ▶ The advantage here is that the positively skewed **innovation term** η_t has a positive probability of being exactly zero in contrast with random component in an **AR(1)** process.
- ▶ Exponentially distributed **random numbers** with means $1/\nu$ and θ are used for generating the times between the **random arrivals** and the magnitudes of the **shots** respectively.
- ▶ The so-called **single-shot noise model** is replaced by a **double-shot noise model** if we attempt to maintain **baseflow** and **surface runoff** process separately. **Seasonality** is accounted for by using different set of parameters for each calendar month.

Models for daily flows: problems and references

- ▶ The main **shortcomings** are on account of undesirable fluctuations on the **recessions** with faster **decay rates** and excessive carry-over effects between months, and biased parameters.
- ▶ Also, **coefficient of skewness** in generated data tend to be higher than historic data.
- ▶ Another inadequacy is the basic assumption of **independence** in **storm events**.
- ▶ **Daily flow models** must also cope with the **dependency** in rainfall inputs, which could be approximated by **Markov chains**. This can be useful in **intermittent rivers** where daily flow can be zero some days and the **ascension** and **recession** behaviour is generally sharp.
- ▶ In the case of **perennial rivers**, usually large river, some deterministic aspects of river flow needs to be incorporated in daily flow models.